AMAN SHASTRY

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EDUCATION

University of Michigan (GPA: 4.00) **Ann Arbor, MI**
Bachelor of Science in Engineering in Data Science, Minor in Mathematics *May 2028*
Coursework: Machine Learning, Web Systems, Applied Regression Analysis, Data Structures and Algorithms, Statistics and AI, Applied Linear Algebra, Multivariable Calculus

SKILLS

Languages: Python, C++, SQL, Dart
ML & Data: PyTorch, Hugging Face Transformers, TensorFlow, NumPy, SciPy, SymPy, Pandas
Tools & Platforms: Git, Linux, IBM Cloud (PowerVS), Flutter

RESEARCH

Independent Researcher **Remote**
"Beyond First-Order Task Vector Superposition: Orthogonal Residuals in In-Context Learning" *June 2026 – Present*
- Showed that hidden states of 5 instruction-tuned LLMs (Llama-3 1B–8B, Gemma-2-2B, Qwen-2.5-3B) under mixed in-context tasks follow ratio-weighted linear superposition, beating all 720 permutations of an exact null at every layer
- Discovered an orthogonal residual that first-order task-vector models miss; it grows with depth and is most resolvable above a split-half noise floor in middle layers (0.38–0.56 relative depth)
- Causally validated the residual on Llama-3.2-3B: ablating it shifted outputs 4.7–8.0× more (KL divergence) than the strongest of 30 norm-matched random ablations, winning in all 18 mixed-task settings
- Ran the project end-to-end independently in PyTorch across 3 task families (arithmetic, factual recall, MMLU)

WORK EXPERIENCE

IBM **New York, NY**
Customer Success Engineer Co-op *August 2026 – Present*
- Built a hybrid-cloud backup demo using IBM Power Virtual Server (PowerVS) and Cobalt Iron Compass on AIX, positioning an enterprise-grade backup solution for small and medium businesses
- Exploring AI applications on IBM Power and IBM Cloud, from core technical components to the sales plays used to position them with clients
- Completing technical enablement in IBM Cloud infrastructure and PowerVS to support presales engagements for enterprise clients

PROJECT EXPERIENCE

Scientific Computation Model Context Protocol Server **Remote**
Independent Project *June 2025*
- Built and deployed an MCP server exposing 26 tools that let LLMs compute exact linear algebra and vector calculus results (eigendecomposition, SVD, QR, gradients, curl, divergence) instead of error-prone in-context math
- Scaled to 25,000+ monthly tool calls with a 90%+ success rate; built in Python (NumPy, SymPy), deployed on Smithery

Michigan Data Science Team **Ann Arbor, MI**
Project Lead *January 2025 – Present*
- Co-lead a 10-member team (5 core contributors) detecting Alzheimer's disease from brain MRI; teach CNNs, ResNet fine-tuning, feature visualization, and ensembling through visual explainers and code notebooks
- Built the Vision Transformer for breast cancer histopathology classification (BreakHis); it was the top model at 98.9% accuracy, 0.991 F1, and 0.99 AUC on a held-out test set, ahead of CNN (94.8%) and Bag of Visual Words (85.7%)
- Helped demonstrate that subtype-first training (8-class subtype, then mapped to benign/malignant) improves every model, raising CNN accuracy by 11 points and Bag of Visual Words by 24 points

Google Developer Group on Campus **Ann Arbor, MI**
President *September 2024 – Present*
- Lead a 30-member chapter running 5 concurrent projects with non-profits; launching Google-supported AI workshops and socials this semester
- Led an 8-person team to build and deploy a modular staff- and resident-tracking web app for HomeRise, boosting productivity by digitizing and automating key logistical processes
- Led a 10-person team building an Audio Journal app with a fine-tuned WavLM speech-emotion model (60% classification accuracy), assigning work by members' strengths as needs evolved
- Built a volunteer and foster-pet management web app for the Humane Society of Huron Valley, digitizing manual workflows and cutting administrative overhead by an estimated 30–40%